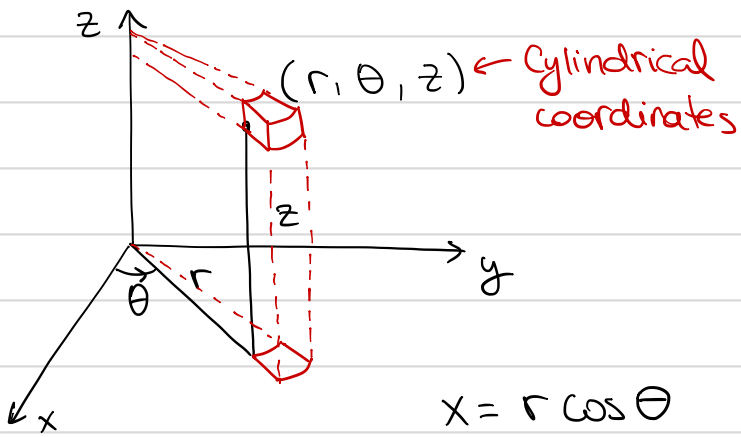
Cylindrical coordinates (§15.7)



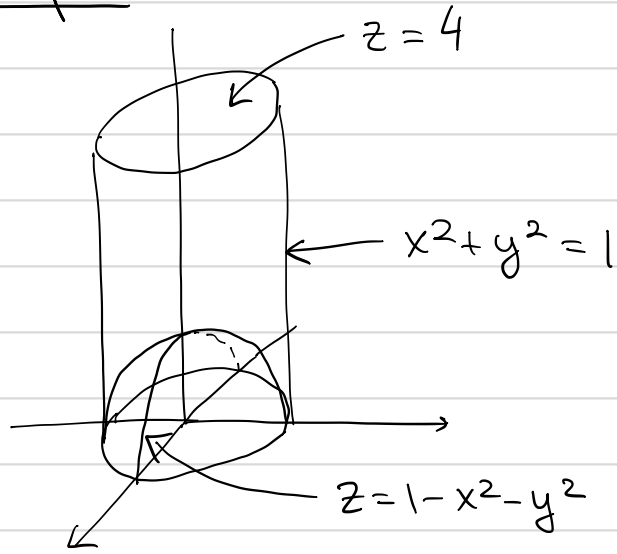
$x = r\cos\theta$

$y = r\sin\theta$

$z = z$

Volume element: $dV = r\,dr\,d\theta\,dz$

Example:



Density: $\rho = Kz$

$$\text{Mass} = \iiint_E \rho\, dV$$

$$= \int_0^{2\pi}\int_0^1\int_{z=1-r^2}^{4} Kz\, dz\, r\,dr\, d\theta$$

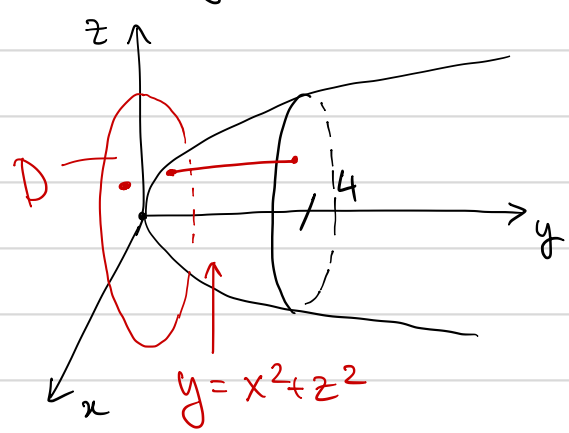
$$= 2\pi K \int_0^1 \underbrace{\frac{1}{2}\left(16-(1-2r^2+r^4)\right)}_{\frac{15}{2}+r^2-\frac{1}{2}r^4} r\,dr$$

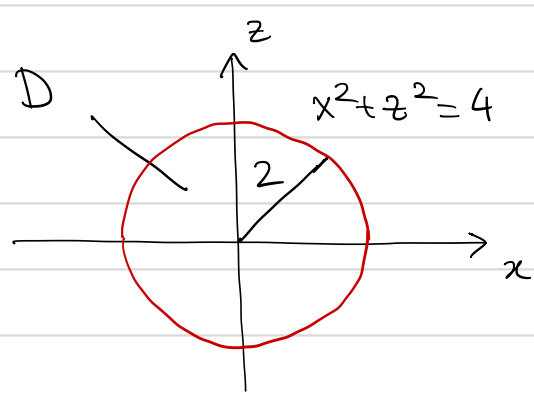
$$= 2\pi K \left[\frac{15}{4}r^2 + \frac{r^3}{3} - \frac{1}{10}r^5\right]_{r=0}^{1}$$

$$= 2\pi K\left(\frac{15}{4}+\frac{1}{3}-\frac{1}{10}\right) //$$

← skip in class

Example: E = region between $y = x^2 + z^2$ and $y = 4$





$$\iiint_E \sqrt{x^2+z^2}\, dV =$$

$x = r\cos\theta$

$z = r\sin\theta$

$y = y$

$$= \iint_D \int_{y=r^2}^{4} r \, dy \, dA$$

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^{2} (4-r^2)\, r \cdot r \, dr \, d\theta$$

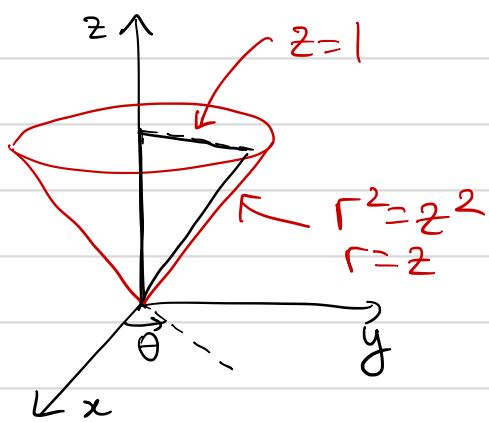
$$= 2\pi \left[\frac{4r^3}{3} - \frac{r^5}{5} \right]_0^2 = 2\pi \left[\frac{32}{3} - \frac{32}{5} \right]$$

$$= \frac{128}{15}\pi \;//$$

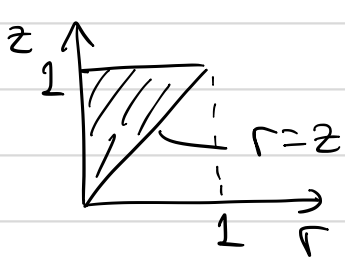
<u>Example</u>: Moment of inertia about z-axis of solid cone bounded by

$x^2+y^2=z^2$ and $z=1$.

Uniform density: $\rho = 1$.



θ-slice:



$$I_z = \iiint_E (x^2+y^2)\, \rho \, dV$$

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^{1} \int_{z=r}^{1} r^2 \cdot 1 \cdot r \, dz \, dr \, d\theta$$

$$= 2\pi \int_0^1 r^3 (1-r)\, dr$$

$$= 2\pi \left(\frac{r^4}{4} - \frac{r^5}{5} \right) \Big|_0^1$$

$$= 2\pi \left(\frac{1}{4} - \frac{1}{5} \right) = \frac{\pi}{10} \;//$$

<u>Example</u>: Volume between:

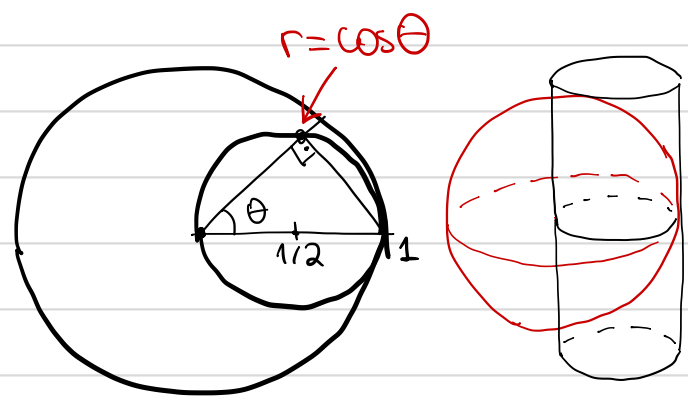
$x^2+y^2+z^2=1$ (sphere)

$r = \cos\theta$ (cylinder)

$$\left.\begin{aligned} r^2 &= r\cos\theta \\ x^2+y^2 &= x \\ (x-\tfrac{1}{2})^2 + y^2 &= \tfrac{1}{4} \end{aligned}\right\}$$

Center: $(\frac{1}{2}, 0)$

Radius: $\frac{1}{2}$

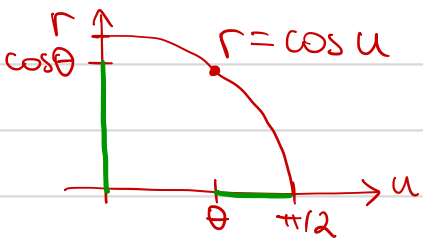


$$\int_{\theta=-\frac{\pi}{2}}^{\pi/2}\int_{r=0}^{\cos\theta}\int_{z=-\sqrt{1-r^2}}^{\sqrt{1-r^2}} dz\, dr\, d\theta$$

$$=\int_{-\pi/2}^{\pi/2}\int_{0}^{\cos\theta} 2\sqrt{1-r^2}\, dr\, d\theta$$

$r=\cos u \longrightarrow dr=-\sin u\, du$

$$=\int_{-\pi/2}^{\pi/2}\int_{\theta}^{\pi/2} 2\underbrace{\sqrt{\sin^2 u}}_{\sin(u)\ \text{since}\ 0\le u\le\pi}(-\sin u)\, du\, d\theta$$



$\cdots = \frac{\pi}{2}(\pi-1)$